

```
import pandas as pd
import datasets
```

```
sciq = pd.read_csv('/content/sciq_clean.csv')
arc = pd.read_csv('/content/arc_clean.csv')
mmlu = pd.read_csv('/content/mmlu_clean.csv')
```

```
combined_df = pd.concat([sciq, arc, mmlu], ignore_index=True)
display(combined_df.head())
```

	dataset_source	text	answer_key
0	SciQ	Question: When atoms gain or lose electrons, t...	D
1	SciQ	Question: What control reproductive cycles, wh...	B
2	SciQ	Question: Wich disorder arises when defective ...	B
3	SciQ	Question: Because core electrons are closer to...	B
4	SciQ	Question: "endocrine" and "exocrine" are two t...	D

```
combined_df= combined_df.sample(frac=1, random_state=42).reset_index(drop=True)
display(combined_df.head())
```

	dataset_source	text	answer_key
0	SciQ	Question: The joints of the carpal bones in th...	B
1	MMLU	Question: Trish's resting heart rate is 50 bea...	B
2	MMLU	Question: A newspaper company sold 179,912 new...	D
3	SciQ	Question: Tuna have been shown to contain high...	C
4	MMLU	Question: Identify the Brønsted-Lowry conjugat...	C

```
combined_df.shape
```

```
(5116, 3)
```

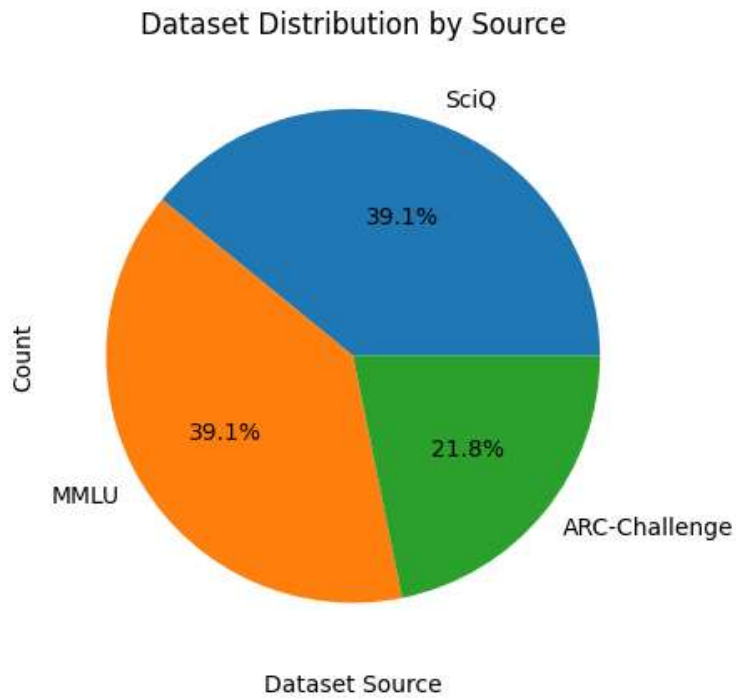
```
combined_df.describe()
```

	dataset_source	text	answer_key
count	5116	5116	5116
unique	3	5109	4
top	SciQ	Question: A rod measures 1.00 m in its rest sy...	D
freq	2000	2	1316

```
import matplotlib.pyplot as plt
import seaborn as sns
```

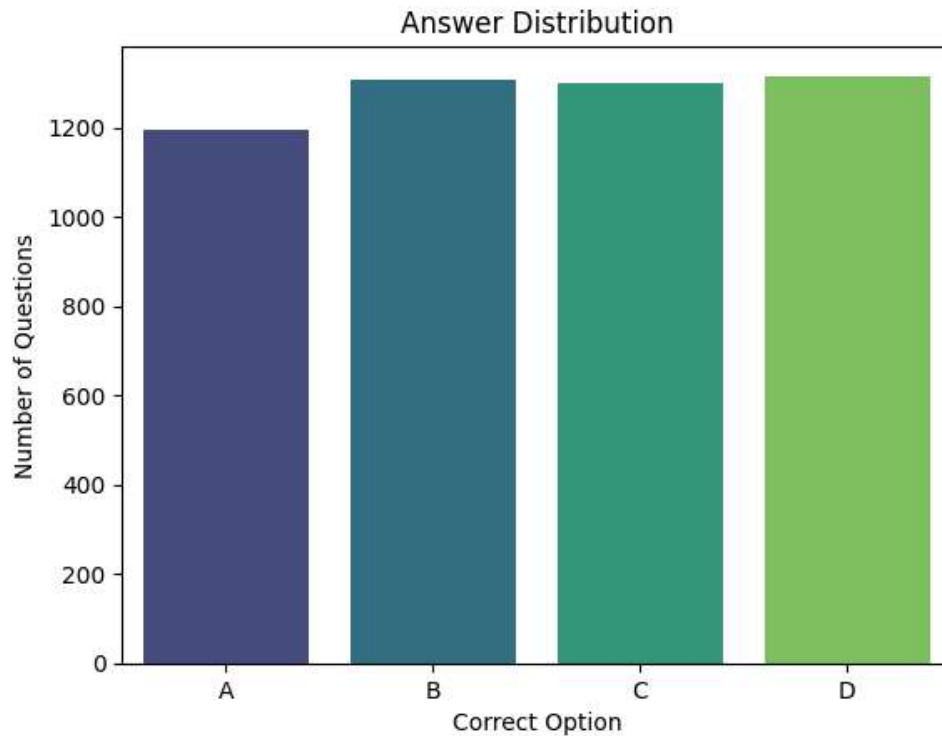
```
combined_df['dataset_source'].value_counts().plot.pie(autopct='%1.1f%%')
plt.xticks(rotation=90)
plt.title('Dataset Distribution by Source')
```

```
plt.xlabel('Dataset Source')
plt.ylabel('Count')
plt.show()
```



```
import warnings
warnings.filterwarnings('ignore')
```

```
sns.countplot(data=combined_df, x='answer_key', order=['A', 'B', 'C', 'D'], palette='viridis')
plt.title('Answer Distribution')
plt.xlabel('Correct Option')
plt.ylabel('Number of Questions')
plt.show()
```



```
combined_df.to_csv('combined_data.csv')
```

```
train_df = combined_df.sample(frac=0.8, random_state=42)
test_df = combined_df.drop(train_df.index)
```

```
from datasets import Dataset
hf_train = Dataset.from_pandas(train_df)
hf_test = Dataset.from_pandas(test_df)
print("Hugging Face Train Dataset:")
display(hf_train[0])
print("\nHugging Face Test Dataset:")
display(hf_test[0])
```

```
Hugging Face Train Dataset:
{'dataset_source': 'MMLU',
 'text': 'Question: Find the quotient of  $2.16 \div 0.06$ .\n\nA. 3.6\nB. 36\nC. 0.01272\nD. 12.72\n\nAnswer: B',
 'answer_key': 'B',
 '__index_level_0__': 530}
```

```
Hugging Face Test Dataset:
{'dataset_source': 'SciQ',
 'text': 'Question: Tuna have been shown to contain high levels of what metal?\n\nA. titanium\nB. cadmium\nC. mercury\nD. iron\n\nAnswer: C',
 'answer_key': 'C',
 '__index_level_0__': 3}
```

```
import os
os.environ["HF_HUB_DISABLE_PROGRESS_BARS"] = "1"
```

```
!pip install -q -U transformers accelerate bitsandbytes
```

```
import os
os.environ['LD_LIBRARY_PATH'] += ':/usr/local/cuda-13.0/lib64'
```

```
from google.colab import userdata
from huggingface_hub import login
```

```
hf_token = userdata.get('HF_TOKEN')
login(hf_token)
print("Authenticated successfully!")
```

Authenticated successfully!

```
import torch
from transformers import AutoTokenizer, AutoModelForCausalLM, BitsAndBytesConfig
model_id = "google/gemma-2b"
bnb_config = BitsAndBytesConfig(
    load_in_4bit = True,
    bnb_4bit_quant_type = "nf4", # a specific math algo for qlora.
    bnb_4bit_compute_dtype=torch.float16
)
```

```
tokenizer = AutoTokenizer.from_pretrained(model_id)
```

```
print("Downloading Gemma 2B")
model = AutoModelForCausalLM.from_pretrained(model_id, quantization_config=bnb_config, device_map=
print("Sucess! The engine is running")
```

Downloading Gemma 2B

Loading weights: 100% 164/164 [00:01<00:00, 95.42it/s]

Sucess! The engine is running

```
import warnings
warnings.filterwarnings('ignore')
```

```
from transformers import pipeline
generator = pipeline("text-generation", model=model, tokenizer=tokenizer)
prompt = """"Question: What is the main function of red blood cells?
Answer: """"
result = generator(prompt, max_new_tokens=50)
print("\n--- AI TEST RESPONSE ---")
print(result[0]['generated_text'])
```

[transformers] Both `max\_new\_tokens` (=50) and `max\_length` (=20) seem to have been set. `max_new_to

--- AI TEST RESPONSE ---

Question: What is the main function of red blood cells?

Answer:

Red blood cells are the main component of the blood. They have a biconcave shape so that they can e

```
correct_answers = 0
total_questions = 100
print(f"Testing baseline Gemma 2B on {total_questions} questions...")
```

Testing baseline Gemma 2B on 100 questions...

```
import re
```

```
for i in range(total_questions):
    raw_text = hf_test[i]['text']
    question_only = raw_text.split("Answer:")[0] + "Answer:"
    question_text = question_only.strip()
    correct_answer = hf_test[i]['answer_key'].strip().upper()
    inputs = tokenizer(question_text, return_tensors="pt").to(model.device)
    with torch.no_grad():
        outputs = model.generate(**inputs, max_new_tokens=2, pad_token_id=tokenizer.eos_token_id,
        decoded = tokenizer.decode(outputs[0], skip_special_tokens=True)
        clean_text = decoded.split("Answer:")[-1].upper().replace("OPTION", "")
        match = re.search(r'[ABCD]', clean_text)
        last_letter = match.group(0) if match else "X"
        if last_letter == correct_answer:
            correct_answers += 1
    if (i + 1) % 10 == 0:
        print(f"Graded {i+1}/{total_questions} questions...")
```

```
Graded 10/100 questions...
Graded 20/100 questions...
Graded 30/100 questions...
Graded 40/100 questions...
Graded 50/100 questions...
Graded 60/100 questions...
Graded 70/100 questions...
Graded 80/100 questions...
Graded 90/100 questions...
Graded 100/100 questions...
```

```
baseline_accuracy = (correct_answers / total_questions) * 100
print("BASELINE GEMMA 2B RESULTS")
print(f"\tCorrect : {correct_answers} / {total_questions}")
print(f"\tAccuracy: {baseline_accuracy:.2f}%")
```

```
BASELINE GEMMA 2B RESULTS
    Correct : 39 / 100
    Accuracy: 39.00%
```

```
from google.colab import drive
drive.mount('/content/drive')
import json
path = "/content/drive/MyDrive/Colab Notebooks/start1.ipynb"
with open(path, "r", encoding="utf-8") as f:
    nb = json.load(f)
nb.get("metadata", {}).pop("widgets", None)
for cell in nb.get("cells", []):
    cell.get("metadata", {}).pop("widgets", None)
with open(path, "w", encoding="utf-8") as f:
    json.dump(nb, f, ensure_ascii=False, indent=1)
print("Fixed notebook metadata!")
```